Copula-Based Modeling of High-Dimensional Medical Tabular Data

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Abstract

This technical report presents a comprehensive framework for modeling high-dimensional medical tabular data using copulas. Our data set consists of approximately 10 clinical variables and 2000 peptides, with categorical and numerical features. We describe our approach to preprocessing, including handling categorical variables and missing values, marginal distribution estimation, and copula modeling with a focus on Gaussian copulas. We also explore non-Gaussian copulas and their potential advantages for capturing tail dependencies in medical data. Our methodology enables the generation of synthetic data that preserve both the marginal distributions and complex dependency structures present in the original dataset.

1 Copula Modeling Framework

1.1 Introduction to Copulas

Copulas provide a powerful mathematical framework for modeling multivariate distributions by separating the dependence structure from the marginal distributions. This separation offers significant flexibility when modeling complex, high-dimensional data such as our medical dataset comprising clinical variables and peptide measurements.

Formally, a copula $C : [0, 1]^d \to [0, 1]$ is a multivariate cumulative distribution function with uniform marginals on $[0, 1]$. For a random vector $\mathbf{X} = (X_1, \ldots, X_d)$ with joint distribution function F and marginal distributions $F_1, \ldots, F_d$, Sklar's theorem states that there exists a copula C such that:

$$F(x_1, \ldots, x_d) = C(F_1(x_1), \ldots, F_d(x_d))$$

In the case where all marginal distributions F_i are continuous, the copula C is uniquely determined.

Intuitively, copulas allow us to model the dependencies between variables independently of their individual behaviors. This property is particularly valuable for our heterogeneous medical dataset, where clinical variables and peptide measurements follow different distributions and exhibit complex interdependencies.

1.1.1 Suitability for High-Dimensional Data

Copulas offer several key advantages that make them particularly well-suited for modeling high-dimensional data such as our medical dataset:

1. **Flexibility with Heterogeneous Data**: Our medical dataset contains variables of different types (binary clinical indicators, continuous measurements) and distributions (normal, right-skewed, zero-inflated). Copulas naturally accommodate this heterogeneity by allowing different marginal distributions for each variable while modeling their dependencies in a unified framework.

2. **Scalability**: Copula models scale relatively well to high dimensions,, as they do not require a large amount of data and are quite computationally efficient. As copulas separate the estimation of marginals from the estimation of the dependence structure, we can estimate marginals in parallel, and the primary computational challenge lies in calculating the correlation matrix, which is feasible in our case.

1.2 Data Preprocessing for Copula Modeling

Before fitting copulas, several preprocessing steps were necessary to address the specific characteristics of our dataset:

1.2.1 Handling Categorical Variables

For categorical variables, we employed a continuous transformation approach:

1. For each categorical variable with k categories, we divided the interval $[0, 1]$ into k segments, with segment widths proportional to the observed class frequencies.

2. Each categorical value was replaced by a random sample from a uniform distribution on its corresponding segment.

For example, for a binary variable with 70% of values being 0 and 30

- Category 0: Uniform random values in $[0, 0.7]$

- Category 1: Uniform random values in $(0.7, 1]$

This approach preserves the distributional properties while creating continuous representations suitable for copula modeling. It is worth noting that this method introduces an arbitrary ordering for categorical variables with more than two categories. However, since our clinical categorical variables are all binary, this limitation does not affect our analysis. For multi-category variables, one-hot encoding would be preferable.

During inference, the generated continuous values are mapped back to categorical values on the basis of which segment they fall into.

1.2.2 Handling Missing Values

Missing values were addressed using a two-component approach:

1. For each variable with missing values, we created a binary indicator variable marking the presence (1 – observed) or absence (0 – missing) of data.

2. The missing values in the original variable were imputed using random samples from the observed values of that variable.

The indicator variables were then treated as categorical variables following the procedure described above. This approach enables the copula model to capture the patterns of missingness, which may contain important information. The imputation strategy does not modify the marginal distribution, while the indicator variable allows for learning of dependency patterns.

1.3 Estimation of Marginal Distributions

For each continuous variable (including transformed categorical variables), we fit multiple probability distributions and select the best fit based on a composite criterion combining:

1. Kullback-Leibler (KL) divergence between the empirical and fitted distributions

2. Akaike Information Criterion (AIC)

3. Bayesian Information Criterion (BIC)

The candidate distributions included:

- Normal

- Gamma

- Beta
- Log-normal
- Exponential
- Student's t
- Uniform
- Truncated normal

The selected marginal distributions F_i were then used to transform the data to the uniform scale $[0, 1]$ by the integral probability transform: $U_i = F_i(X_i)$.

1.4 Gaussian Copula Estimation

For the copula structure, we primarily employed Gaussian copulas because of their computational tractability and flexibility for high-dimensional data. The Gaussian copula is defined as:

$$C_{\mathbf{R}}(u_1, \ldots, u_d) = \Phi_{\mathbf{R}}(\Phi^{-1}(u_1), \ldots, \Phi^{-1}(u_d))$$

where $\Phi_{\mathbf{R}}$ is the joint cumulative distribution function of a multivariate normal distribution with correlation matrix $\mathbf{R}$, and Φ^{-1} is the inverse cumulative distribution function of the standard normal distribution.

The key challenge in our high-dimensional setting ($\sim$2000 peptides plus clinical variables) was estimating a valid correlation matrix $\mathbf{R}$. We employed Higham's algorithm [**?**] to ensure that our estimated correlation matrix was positive semi-definite (PSD), which is necessary for a valid Gaussian copula.

Higham's algorithm iteratively projects an initially estimated correlation matrix onto the space of positive semi-definite matrices while preserving the diagonal elements (which remain 1). The algorithm minimizes the Frobenius norm between the original matrix and the PSD approximation.

1.5 Model Fitting and Data Generation

The complete copula model fitting procedure consisted of:

1. Transforming the original data to continuous representations as described in Section 1.2.
2. Estimating marginal distributions F_i for each variable.
3. Transforming the data to the uniform scale using the probability integral transform: $U_i = F_i(X_i)$.
4. Converting the uniform data to normal scores using the inverse normal CDF: $Z_i = \Phi^{-1}(U_i)$.
5. Estimating the correlation matrix $\mathbf{R}$ from these normal scores Z_i.
6. Applying Higham's algorithm to $\mathbf{R}$ to ensure positive semi-definiteness, yielding $\mathbf{R}_{PSD}$.

To generate synthetic samples from the fitted copula model, we:

1. Generated samples $\mathbf{z} = (z_1, \ldots, z_d)$ from a multivariate normal distribution $\mathcal{N}(0, \mathbf{R}_{PSD})$.
2. Transformed the normal samples to uniform samples using the normal CDF: $u_i = \Phi(z_i)$.
3. Transformed the uniform samples $\mathbf{u} = (u_1, \ldots, u_d)$ to the original scale using the inverse CDFs of the fitted marginal distributions: $x'_i = F_i^{-1}(u_i)$.
4. Converted continuous representations x'_i back to categorical values where appropriate (for original categorical variables and missing indicators).
5. Reintroduced missing values through multiplying each variable with its missing indicator variable.

This procedure produces synthetic data that preserves both the marginal distributions of individual variables and their complex dependency structure, making it suitable for various downstream applications in medical data analysis.

1.6 Fidelity Results

To assess the fidelity of the generated synthetic data with respect to the original data, we performed statistical comparisons on a per-variable basis. Specifically, for each peptide generated by the copulas, we conducted a Mann-Whitney U test to compare the distribution of the original data with the distribution of the synthetic data.

The results of this analysis indicated a high degree of distributional similarity. Across all peptides tested, only 0.98 percent were found to have statistically significant differences (p-value < 0.05 after appropriate correction for multiple comparisons) between their original and synthetic distributions. This low percentage suggests that the copula-based generation process largely succeeds in capturing and reproducing the individual characteristics of the variables.

A Exploration of Non-Gaussian Copulae

Alongside improving the marginal distributions, we explored enhancements to the copula distribution itself. While the Gaussian copula served as our primary model due to its simplicity and standard usage, we investigated alternative copula families that might better capture certain dependency structures present in our medical data.

The key concept guiding our exploration was **tail dependence**, which measures the likelihood of observing extreme values in one variable given extreme values in another. Mathematically, for two random variables X and Y with marginal distribution functions F_X and F_Y, the upper tail dependence coefficient λ_U is defined as:

$$\lambda_U = \lim_{u \to 1^-} P(Y > F_Y^{-1}(u) | X > F_X^{-1}(u)) = \lim_{u \to 1^-} \frac{1 - 2u + C(u,u)}{1-u}$$

Similarly, the lower tail dependence coefficient λ_L is:

$$\lambda_L = \lim_{u \to 0^+} P(Y \leq F_Y^{-1}(u) | X \leq F_X^{-1}(u)) = \lim_{u \to 0^+} \frac{C(u,u)}{u}$$

(provided the limits exist).

The Gaussian copula exhibits no tail dependence ($\lambda_U = \lambda_L = 0$, for correlation $|\rho| < 1$), meaning it may underestimate the probability of simultaneous extreme events. In our peptide data, this could be particularly problematic if certain peptides tend to have coordinated extreme abundance values.

We explored several alternative copula families with varying tail dependence properties:

1. **t-copula**: Derived from the multivariate Student's t-distribution, this copula exhibits symmetric tail dependence, with both $\lambda_U > 0$ and $\lambda_L > 0$. The degree of tail dependence is controlled by the degrees of freedom parameter ν.

2. **Archimedean copulas**: This class includes several families:
   - *Clayton copula*: Exhibits lower tail dependence ($\lambda_L > 0, \lambda_U = 0$).
   - *Gumbel copula*: Exhibits upper tail dependence ($\lambda_L = 0, \lambda_U > 0$).
   - *Frank copula*: Has no tail dependence ($\lambda_L = \lambda_U = 0$) but possesses a different dependence structure compared to the Gaussian copula.

3. **Vine copulas** (Pair-Copula Constructions): These hierarchical structures combine bivariate copulas (which can be from different families) to model complex dependencies in high dimensions, offering great flexibility but increased complexity.

In this work we focus on comparing the **t-copula** to the **Gaussian copula**, and leave the rest for future work. For preliminary evaluations, we measured tail dependence using an approximation of the limit from the definition by evaluating the conditional probability at a specific high quantile (e.g., $u = 0.95$).

Figure 1 illustrates that t-copulas potentially provide a better estimate of empirical tail dependence compared to Gaussian copulas in some scenarios, although perfect estimation remains challenging. Our

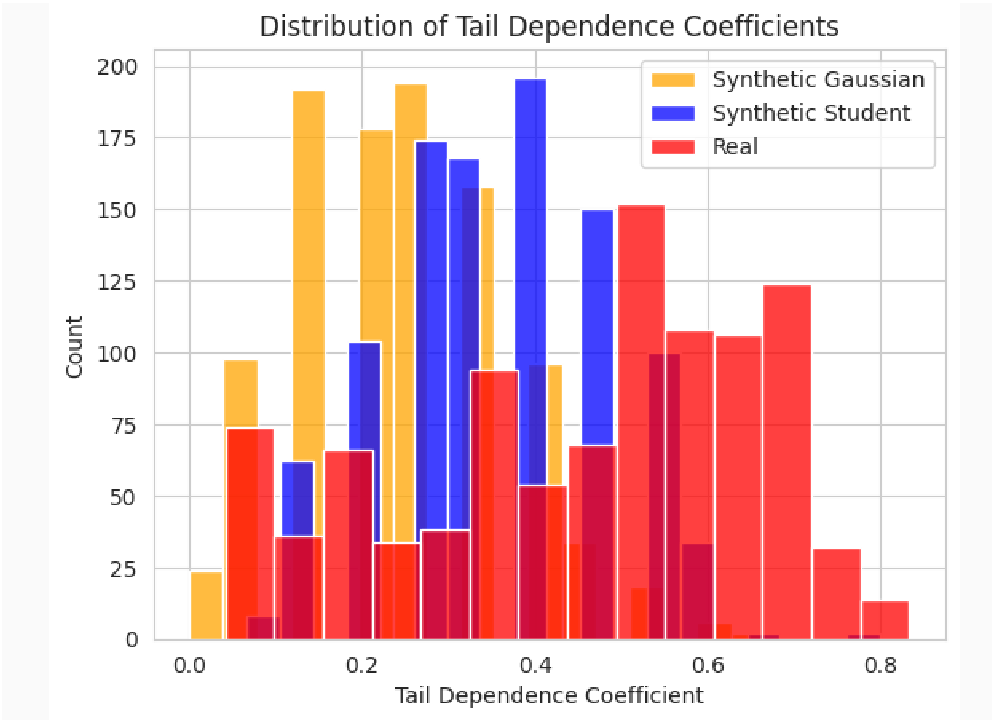

Figure 1: Comparison of empirical vs. estimated tail dependence for Gaussian and t-copulas.

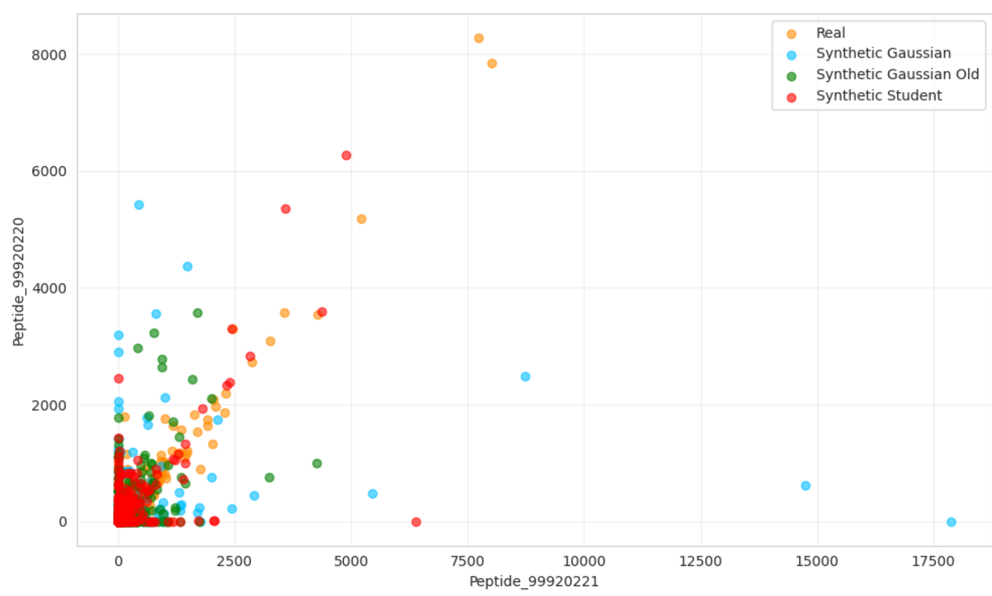

Figure 2: Example of a peptide with strong tail dependence - a comparison of copulas. Student copula is show to better capture the codependence in the distribution tails.

preliminary evaluations suggest that alternative copulas produce synthetic data that is comparable to that generated by Gaussian copulas, without clear superiority or inferiority based on global metrics. The differences appear subtle and may be more relevant for specific downstream applications that are sensitive to extreme value co-occurrences.

Further investigation is needed to:

1. Develop robust measures of tail dependence suitable for high-dimensional, potentially sparse data.
2. Assess the impact of different copula choices on specific downstream predictive tasks using appropriate evaluation metrics.
3. Explore computational optimizations and fitting strategies for non-Gaussian and vine copulas in high dimensions.